

P5 Planning, analysis and evaluation

Practical · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Start with a biological hypothesis and testable prediction, then design sampling to reduce bias and enough repeats to estimate variability. Differentiate correlation from causation; choose a suitable graph/statistical test and interpret it against assumptions. Improvements should respond to limits such as uncontrolled temperature or small sample size.

Worked example: follow the reasoning

A strong temperature–enzyme correlation can be confounded by pH changes; use a buffer and repeat at each temperature to distinguish effects.

Quick check · multiple choice

A What would strengthen a proposed causal explanation?

[1]

- A Controlled intervention and repeats
- B A larger graph title
- C One unreplicated observation
- D Removing units

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A study finds leaf area and transpiration rate correlated. Explain two reasons this does not alone establish causation and propose one improvement.

[3]

Final answer: _____

Answers and explanation

Quick check: A. Controlled intervention and repeats

Question 1 [3 marks]

Other factors such as stomatal density or temperature may covary [1]; correlation gives association not controlled intervention [1]; hold conditions constant/normalise per unit area and replicate [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.